Sinusoidal Distribution

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Abstract

We introduce the *Sinusoidal distribution*, a flexible 4-parameter location-scale-skewness-kurtosis family of distributions, based on Gilbert's Sine distribution (1893). The mathematical properties of this distribution are studied. Examples of fitting the distribution to various statistical objects are considered to demonstrate the distribution's flexibility. Finally, some future avenues are considered for exploration.

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1 Introduction

There is significant need for flexible probability distributions in statistical domains. Although most frequently used densities are unimodal and "bell shaped", their shapes can be arbitrarily complex. Several new candidates, such as Chalabi, Scott, and Wuertz 2012, and at the same time, multiple new techniques, some of which are summarized in Anusha, Bhatracharyulu, and Jayasree 2024, have been devised to generate more flexible shapes from existing distributions.

One instinctive choice for a bell-shape is the crest of a sine-curve. Even disregarding its elegant mathematical properties, this crest forms a visually natural candidate as a probability density function. Gilbert 1893 mentions the *Sinusoidal distribution* in its most well recognized form to study the distribution of moon-craters, although in Edwards 2000 we find a brief history of the same density expression in the works of La Grange 1770 and Pearson 1902.

Several attempts have been made to extend the Sine Distribution into a more flexible array of shapes to model a wide variety of data distributions, including SS transformed Sine distribution (Kumar, U. Singh, and S. K. Singh 2015), Sine-G distribution (Mahmood, Chesneau, and Tahir 2019), and Sine generalized family (Oramulu et al. 2024) although such recent modifications propose formulatically complex extensions of Gilbert's proposed elegant density. In our work, we consider a simple 4-parameter extension of Gilbert's baseline Sine Distribution into a flexible, location-scale-skewness-kurtosis family, that is equally elegant in its density expression to Gilbert's Sine Distribution.

We navigate through the work as follows: at first, we introduce the *Sinusoidal distribution* and investigate its shape properties. We will then take a look at some immediate mathematical results that follow from its density expression. Then we take a deeper look at its mathematical properties using the notion of Generalized Hypergeometric Functions. Then, we provide two instances of the flexibility of the *Sinusoidal distribution* by highlighting its fitting capabilities. Finally, we discuss its possible future applications.

2 The Sinusoidal Distribution

First we introduce the "Standard" *Sinusoidal distribution* family as follows:

Random variable Z is said to follow a Standard *Sinusoidal distribution* when it is distributed according to the density:

$$\boxed{\psi(z;s,k) \equiv \frac{\varsigma(z;s,k)}{A(s,k)} = \frac{1}{A(s,k)}\sin^k(\pi z^s); \quad z \in [0,1]} \tag{1}$$

where the parameters are $s \geq 0, k \geq 0$, the kernel is notated as $\varsigma(z;s,k) := \sin^k(\pi z^s)$, and $A(s,k) := \int_0^1 \sin^k(\pi t^s)\mathrm{d}t$ is the normalizing constant, i.e. the complete integral of the kernel over $[0,1]$. The

notation for the distribution is $Z \sim \text{Sinu}(s, k)$, and naturally, its CDF is given by:

$$\Psi(z; s, k) = \int_0^z \psi(t; s, k)\mathrm{d}t \equiv \frac{A_z(s, k)}{A(s, k)}; \quad z \in [0, 1] \tag{2}$$

where $A_z(s, k)$ denotes the incomplete integral of the kernel.

We now introduce 2 extensions. Firstly, the location scale extension is defined as follows: random variable X follows a location-scale *Sinusoidal distribution* family, notated $X \sim \text{Sinu}(a, d, s, k)$ with parameters $a \in \mathbb{R}, d \geq 0, s \geq 0, k \geq 0$, when X has density:

$$\psi(x; a, d, s, k) = \frac{1}{d}\psi\left(\frac{x-a}{d}; s, k\right); \quad x \in [a, a+d] \tag{3}$$

Next, we introduce a "flipped extension" (for the reason described in subsection 2.1, item 4) to the *Sinusoidal distribution*: random variable X follows a flipped location-scale *Sinusoidal distribution* family, notated $X' \sim \text{Sinu}'(a, d, s, k)$, when $X' :\overset{d}{=} a + d - X$ where $X \sim \text{Sinu}(a, d, s, k)$. Conceivably, X' has density

$$\psi'(x; a, d, s, k) = \frac{1}{d}\psi\left(\frac{a+d-x}{d}; s, k\right) \tag{4}$$

Multivariate Extension

We provide multivariate extension notions of only the standard *Sinusoidal distribution*, i.e. location scale extensions of the multivariate extensions is left implicit.

A: Defined independently using individual (s,k)'s: $\vec{Z}_{m\times 1} \sim \psi_m(\vec{s}_{m\times 1}, \vec{k}_{m\times 1})$:

$$\psi_m(\vec{z}; \vec{s}, \vec{k}) = \prod_{j=1}^{m} \psi(z_j; s_j, k_j)$$

This makes it equivalent to the joint density of m independent Standard Sinusoidal random variables.

B: Defined through a common (s, k) pair via vector generalization

$$\psi_m(\vec{z}; \mathbf{D}, s, k) = \psi(\vec{z}^\intercal \mathbf{D}^{-1}\vec{z}; s, k)$$

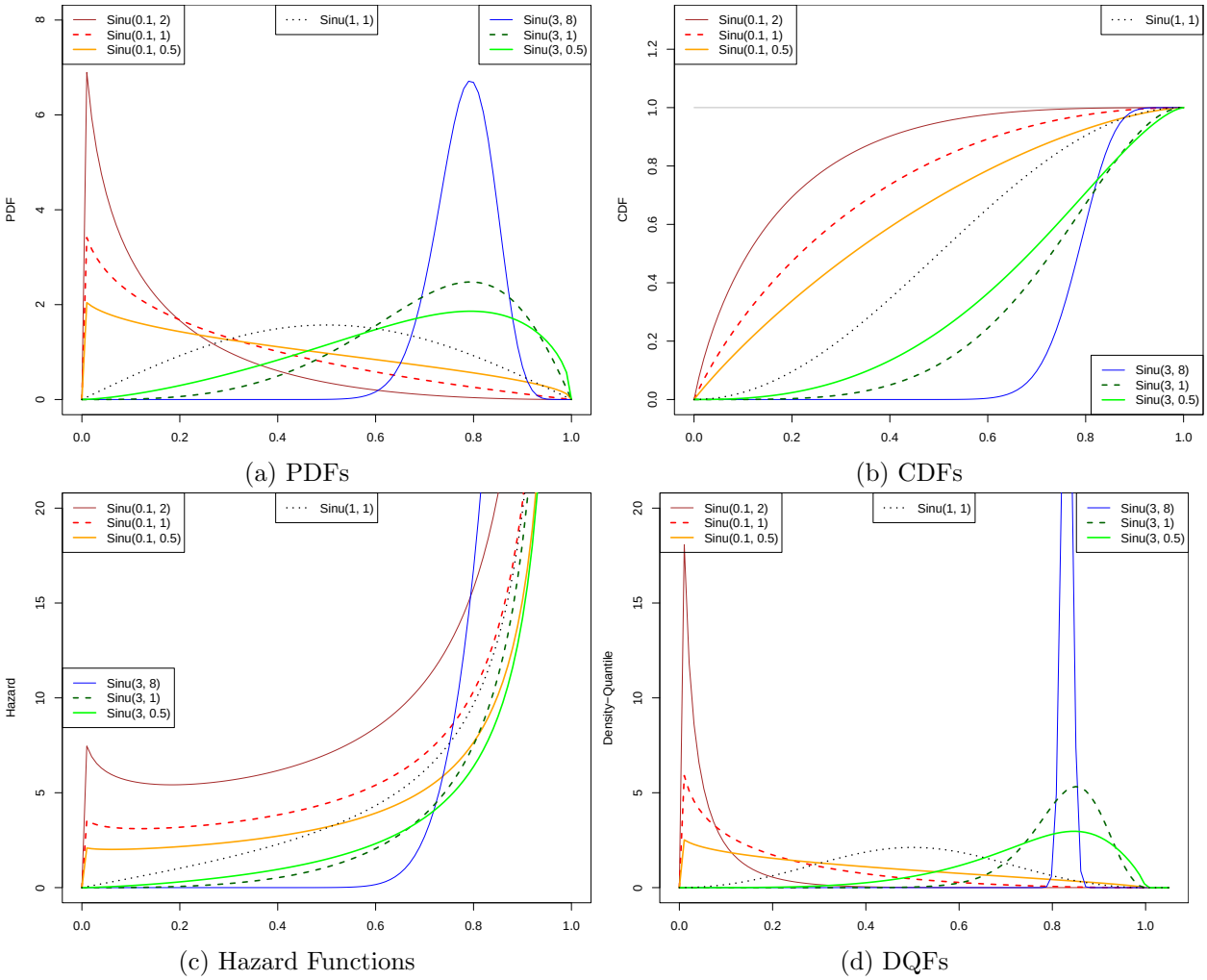
Unsure of the structure of D, what kind of correlation structure D will provide, whether the defined density will be normalized, how s and k will influence its shape here, etc.

C: Defined through the Archimedean copula.

Figure 1 illustrates the PDFs, CDFs, Hazard functions and DQFs (Density-Quantile Functions, as described in Staudte 2017) of some Standard *Sinusoidal distributions*.

2.1 Shape Properties

Some shape properties of the *Sinusoidal distribution* are discussed below:



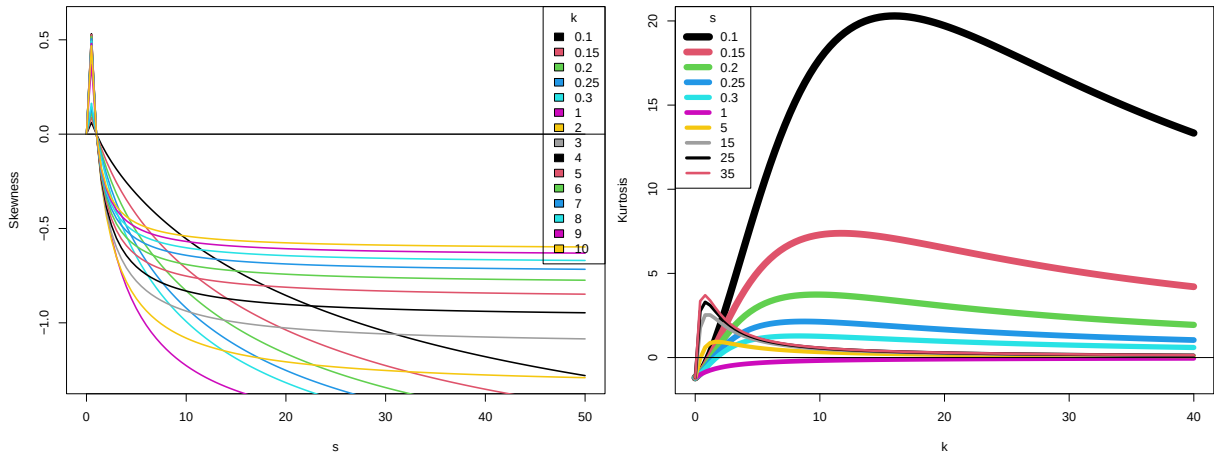
(a) PDFs

(b) CDFs

(c) Hazard Functions

(d) DQFs

Figure 1: Shape plots of different *Sinusoidal distributions*



(a) Skewness vs. s

(b) Kurtosis vs. k

Figure 2: s, k jointly affect skewness & kurtosis

1. Support: $[a, a+d]$. a is the location and d is the scale parameter. Alternative parametrization with $b = a + d$ may be considered.

2. s affects skewness of the *Sinusoidal distribution* as follows:

   (a) $0 < s < 1$: left-leaning (positively/right skewed)

   (b) $s = 1$: symmetric

   (c) $s > 1$: right-leaning (negatively/left skewed)

   . The alternative parametrization $s' = \ln s$ calibrates the parameter choice $s = 0$ to the symmetric family, and on either side of it keeps oppositely signed skewness families.

3. The choice $a = -\frac{d}{2}$, $s = 1$ gives the special symmetric around 0 $\text{Sinu}(-\frac{d}{2}, d, 1, k)$ family, supported on the interval $[-\frac{d}{2}, \frac{d}{2}]$.

4. s (or any transformation of it) does not affect the shape of the distribution symmetrically around the point $s = 1$; i.e. $\nexists s' < 1 : \text{Sinu}(s', k) \overset{d}{=} 1 - \text{Sinu}(s, k) \quad \forall s > 1$. It is for this reason that we introduced the notion of a flipped *Sinusoidal distribution*.

5. k affects kurtosis non-monotonically. See Figure 2.

6. s, k jointly affect both the skewness and the kurtosis. See Figure 2.

7. A $\text{Unif}(a, a + d)$ density results from the *Sinusoidal distribution* density when k=0 or s=0.

8. Degenerate at a for $d = 0$

9. $\text{Sinu}(-\frac{d}{2}, d, 1, k) \xrightarrow[d=\pi\sqrt{k}]{k\to\infty} \mathcal{N}(0, 1)$. See Theorem 3.6.

10. The mode of $\text{Sinu}(s, k)$ is $2^{-1/s}$ and the modal density is $1/A(s, k)$. See Theorem B.1.

3 Preliminary Mathematical Properties

Proofs to theorems mentioned are provided in the Appendix subsection B.1.

Programmatic Notation

We follow these conventions of nomenclature, in both the report and the code as far as possible, to quickly understand the domain of variables, and in some ways also their interpretation.

1. $x \in [a, a+d]$
2. $z :\equiv \frac{x-a}{d} \in [0,1]$
3. $u :\equiv z^s \in [0,1]$ also.
4. $\theta :\equiv \pi u \equiv \pi z^s \in [0, \pi]$
5. $\omega := \theta - \frac{\pi}{2} \equiv \pi z^s - \frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
6. $r = \pm 1$

3.1 Physical Interpretation

Similar to Gilbert 1893 – which imagines intensity (interpreted as probability density) of crater activity on a circular planet as proportional to Gilbert's Sine density $\sin\theta$ – the Standard *Sinusoidal distribution* $\mathrm{Sinu}(s,k)$ can also be physically interpreted as the intensity of crater activity on a (possibly non-circular) planetary surface given by the radial curve:

$$r(z) = r_0 \exp\left(\int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t\right) \tag{5}$$

where the curve $r(z)$ is parametrized by $z \in [0,1]$ such that $\theta = \pi z^s$ is the angle the location $r(z)$ subtends at the origin, and r_0 is the initial radius at $z = 0$. Details are furnished in Remark B.1.

3.2 Transformation

Theorem 3.1 (Expression as a morphed arcsin Beta distribution)**.** *$Y \sim \mathrm{B}(\frac{k+1}{2}, \frac{1}{2})$ and $R \sim \mathrm{Ber}(\frac{1}{2})$, then*

$$Z = \left[R\sin^{-1}\sqrt{Y} + (1-R)(\pi - \sin^{-1}\sqrt{Y})\right]^{\frac{1}{s}}$$

follows a $\mathrm{Sinu}(s,k)$ distribution.

Theorem 3.2 (Joint density expression)**.** *For $X_1 \dots X_n \overset{i.i.d.}{\sim} \mathrm{Sinu}(a,d,s,k)$, their joint density can be expressed as:*

$$f(x_1 \dots x_n) = \frac{1}{A^n(s,k)2^{nk}d^n}\left(\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1}\left[\prod_{j=1}^n r_j\right]\cdot\exp\left[\sum_{j=0}^n \mathrm{i}\pi r_j\left(\frac{x_j-a}{d}\right)^s\right]\right)^k \tag{6}$$

3.3 Concentration Inequalities

Theorem 3.3 (*Sinusoidal distribution* is sub-Gaussian)**.** *$X \sim \mathrm{Sinu}(a, d, s, k)$ is sub-Gaussian with parameter $\frac{d^2}{4}$.*

Proof. $\because a \leq X \leq a + d$ a.s., $\therefore \mathbb{E}\left[\mathrm{e}^{\lambda(X-\mathbb{E}X)}\right] \leq \exp\left(-\frac{\lambda d^2}{8}\right)$ by Hoeffding's lemma.

Note that because it is sub-Gaussian, it is also sub-Exponential. □

3.4 Normalizing constant for the symmetric case

Theorem 3.4 (Normalizing constant, symmetric case)**.**

$$A(1, k) \equiv \int_0^1 \sin^k(\pi t)\mathrm{d}t = \frac{2^k}{\pi} \frac{\Gamma^2\left(\frac{k+1}{2}\right)}{\Gamma(k+1)} \tag{7}$$

$$= \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)} \tag{8}$$

3.5 Incomplete integral of the kernel for the Symmetric case

Theorem 3.5 (Incomplete integral of the kernel for the symmetric case)**.** *When $k \in \mathbb{Z}^+$, $s = 1$, we have –*

$$\int_0^z \sin^k \pi t \, dt = \frac{1}{2^{2n}} \binom{2n}{n} z + \frac{(-1)^n}{\pi 2^{2n-1}} \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} \frac{\sin 2(n-k)\pi z}{2(n-k)} \qquad [k \text{ even}] \tag{9}$$

$$= \frac{1}{\pi 2^{2n}} (-1)^n \sum_{k=0}^{n} (-1)^k \binom{2n+1}{k} \frac{1 - \cos(2n+1-2k)\pi z}{2n+1-2k} \qquad [k \text{ odd}] \tag{10}$$

3.6 Limiting Gaussianity

Theorem 3.6. *For the symmetric about 0* Sinusoidal distribution $\mathrm{Sinu}\left(-\frac{d}{2}, d, 1, k\right)$*, i.e. with $a = -\frac{d}{2}$ and $s = 1$, setting $d \equiv d(k) = \pi\sqrt{k}$ and letting $k \uparrow \infty$, we have:*

$$\psi_k(x) \to \phi(x)$$

where $\phi(x)$ is the PDF of a Standard Normal distribution.

3.7 Limiting Gamma

Theorem 3.7. $\mathrm{Sinu}(a, d, s, k)$ *becomes a Gamma distribution for the parameter choice: ???*

3.8 Other distributions

What about Inverse Gamma, Beta and some other common families?

4 Properties involving the Generalized Hypergeometric Function

For this section, we define Generalized Hypergeometric Function (GHGF) F^p_q, parametrized by $a_1, \ldots, a_p$ and $b_1, \ldots, b_p$, calculated at the argument x as:

$$\mathrm{F}^p_q \left(\begin{matrix} a_1 & \cdots & a_p \\ b_1 & \cdots & b_q \end{matrix} ; x \right) := \sum_{j=0}^{\infty} \frac{(a_1)^{\overline{j}} \cdots (a_p)^{\overline{j}}}{(b_1)^{\overline{j}} \cdots (b_q)^{\overline{j}}} \frac{x^j}{j!}$$

where $(x)^{\overline{n}} := x(x+1)\cdots(x+n-1)$ is the rising factorial / Pochhammer function.

Also define from hereon $\binom{r}{k} = \frac{(r)^{\overline{k}}}{k!}$ as the generalized binomial coefficient.

Some additional notes are provided in Remark B.2. Proofs to the theorems mentioned are provided in subsection B.2.

4.1 Kernel

Theorem 4.1 (Kernel expression via Binomial expansion of Euler form)**.**

$$\varsigma(z; s, k) \equiv \sin^k(\pi z^s) = \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s} \tag{11}$$

Theorem 4.2 (Kernel expression via Generalized Hypergeometric Function)**.**

$$\varsigma(z; s, k) \equiv \sin^k(\pi z^s) = (\pi z^s)^k \left(\mathrm{F}^0_1 \left[\begin{matrix} \cdot \\ \frac{3}{2} \end{matrix} ; -\frac{\pi^2 u^2}{4} \right] \right)^k \tag{12}$$

$$= \mathrm{F}^1_0 \left[\begin{matrix} k \\ \cdot \end{matrix} ; -\exp(-2\mathrm{i}\pi z^s) \right] \tag{13}$$

4.2 CF and MGF

Theorem 4.3 (Characteristic Function and MGF of the Standard *Sinusoidal distribution*)**.** *For $Z \sim \mathrm{Sinu}(s, k)$, its CF is given as –*

$$\varphi_Z(t) = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{m=0}^{\infty} \frac{(\mathrm{i}t)^m}{(m+1)!} \mathrm{F}^1_1 \left[\begin{matrix} \frac{m+1}{s} \\ \frac{m+1}{s} + 1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{14}$$

Additionally, the MGF is given as –

$$\phi_Z(t) \equiv \varphi_Z(-it) = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{m=0}^{\infty} \frac{t^m}{(m+1)!} \mathrm{F}^1_1 \left[\begin{matrix} \frac{m+1}{s} \\ \frac{m+1}{s} + 1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{15}$$

4.3 Moments

Theorem 4.4 (Moments of Standard *Sinusoidal distribution*)**.** *For* $Z \sim \text{Sinu}(s,k)$*,* $\mu_r(s,k) := \mathbb{E}[Z^r]$ *is given by*

$$\mu_r(s,k) = \frac{1}{A(s,k)(2\mathrm{i})^k(r+1)} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \mathrm{F}_1^1 \left[\begin{matrix} \frac{r+1}{s} \\ \frac{r+1}{s}+1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{16}$$

$$= \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} \binom{k}{j} (-1)^j \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+r+1)} \tag{17}$$

4.4 CDF

Theorem 4.5 (CDF of the Standard *Sinusoidal distribution*)**.**

$$\Psi(z;s,k) = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)} z^{ns+1} \tag{18}$$

4.5 Normalizing constant

Theorem 4.6 (Normalizing constant for the Standard *Sinusoidal distribution*)**.**

$$A(s,k) = \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)}$$

5 Numerical and Empirical Properties

5.1 Convolutional approximation

Although the convolution of two iid Sinusoidal RVs is not Sinusoidally distributed, yet a candidate Sinusoidal distribution is often a good approximation of the same. Furthermore, such an approximation can be achieved by prefixing a location-scale readjustment criterion, just like any other family of distributions, i.e.:

$$\text{Sinu}(a,d,s,k) * \text{Sinu}(a,d,s,k) \approx \text{Sinu}(2a,2d,s^*,k^*) \tag{19}$$

for some s^* and k^*.

The following empirical study is conducted in this regard. The sum of 2 iid standard $\text{Sinu}(s,k)$ RVs (i.e. convolutions of their 2 same densities) are considered over a grid of choices of $s,k \in (0,10]$. Every convolution is attempted to be approximated by some candidate $\text{Sinu}(0,2,s^*,k^*)$ by minimizing the Hellinger distance. The results are plotted in the graph below. A poor fit (larger Hellinger distance) is presented as redder on the plot, whereas a good fit (smaller Hellinger distance) appears greener. Candidate s^*,k^* are omitted from display. The results are displayed in Figure 3

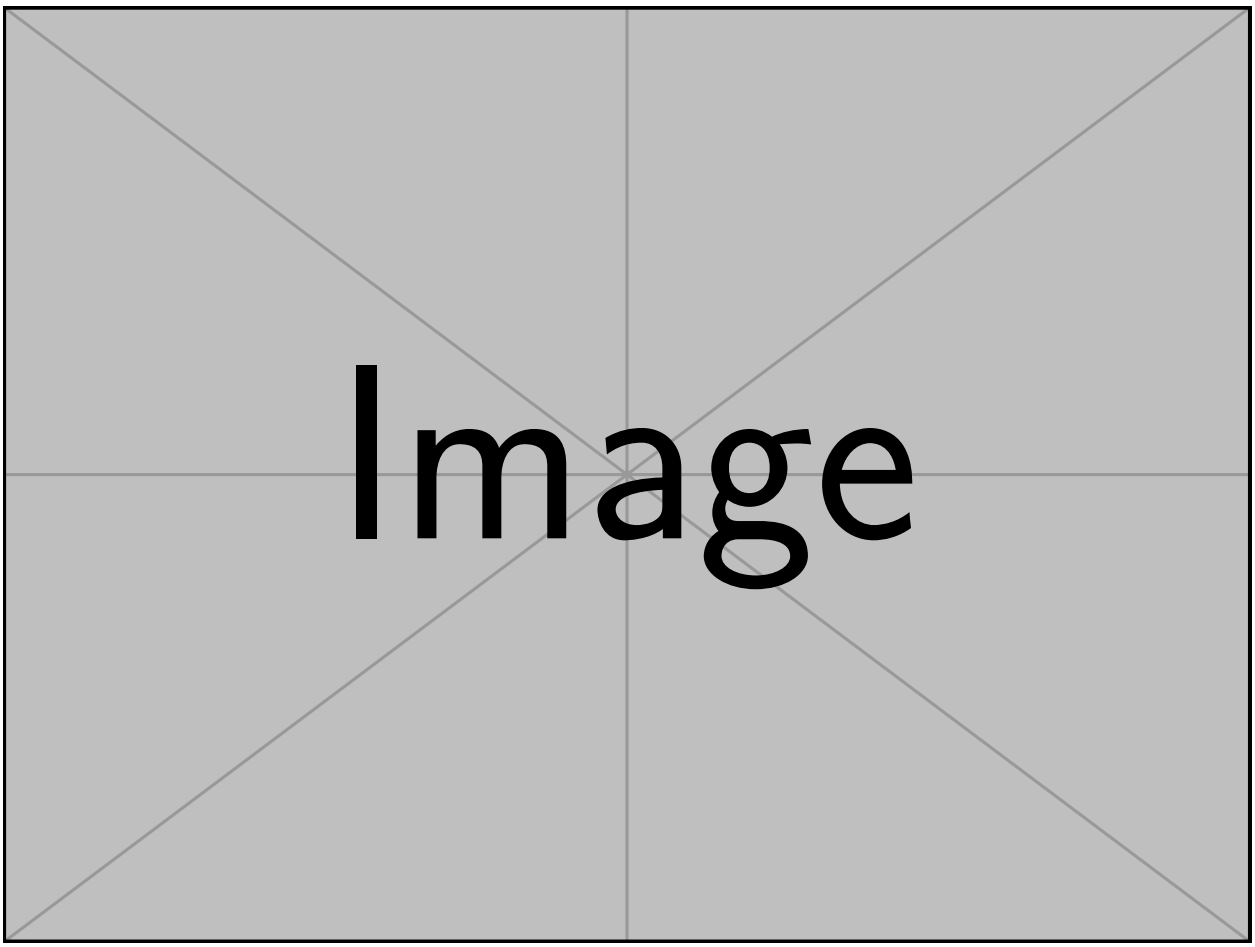


Figure 3: Text

A second study assesses the quality of fits when the location-scale readjustment criterion (Equation 5.1) is lifted. The results are in Figure 4.

6 Fitting

This section discusses fitting the *Sinusoidal distribution* over 2 kinds of statistical objects – MSMs (moment summary measures), and PDFs, in an attempt to demonstrate its flexibility.

It is to be noted that since the *Sinusoidal distribution* is a location-scale family, investigating its fitting flexibility over standardized/normalized statistical objects is good enough, since it can always adapt to location-scale transformed analogues of the aforesaid objects.

6.1 MLE, Score equations

The score equations for maximum likelihood estimation of the parameters of the *Sinusoidal distribution* $\text{Sinu}(a, d, s, k)$ are not analytically solvable, as we detail below. Consider $X_i \overset{\text{i.i.d.}}{\sim} \text{Sinu}(a, d, s, k) \quad \forall i =$

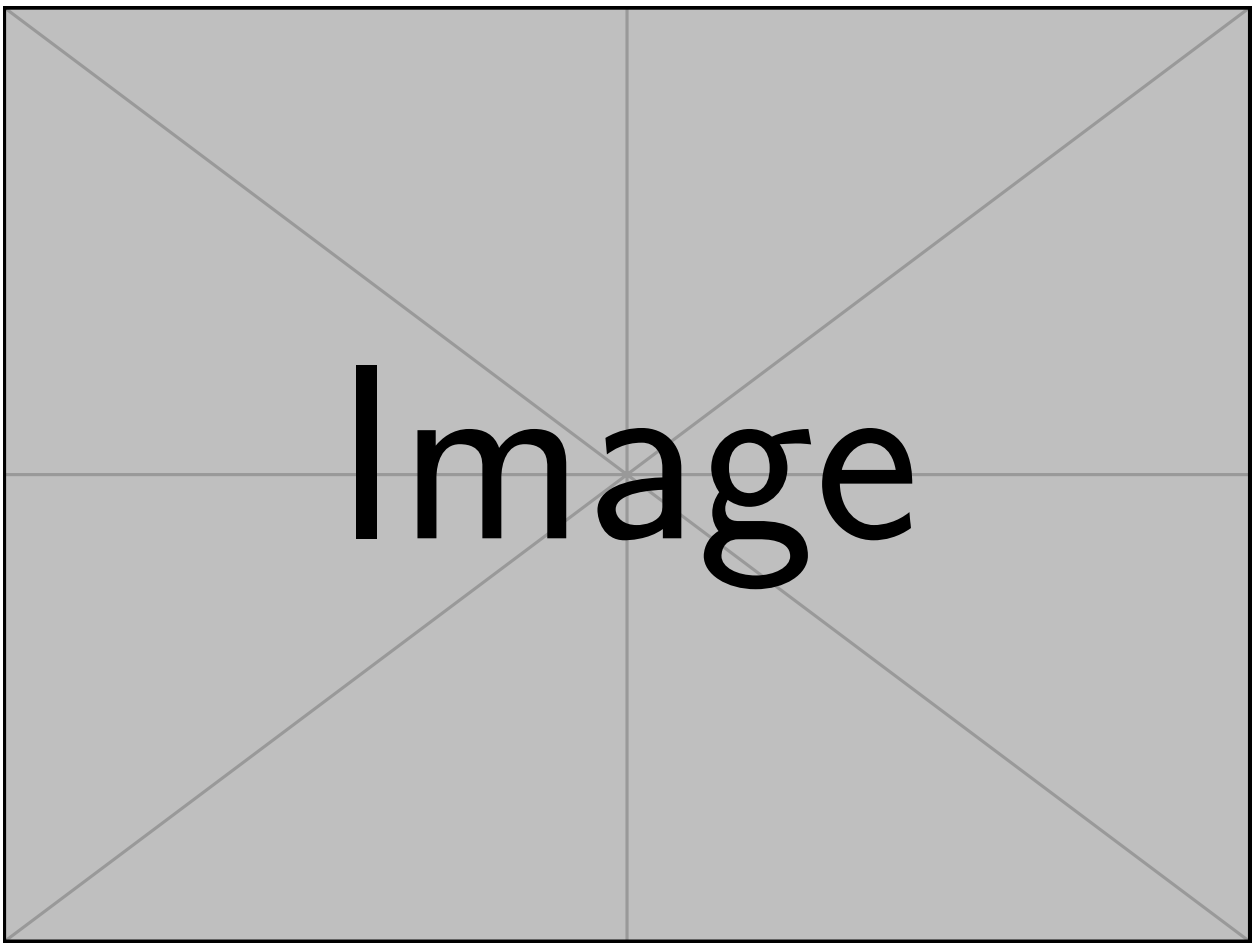


Figure 4: Text

$1(1)n$. The joint log-likelihood is:

$$l(a, d, s, k; \underset{\sim}{x}) = -n \ln(d) - n \ln(A(s, k)) + k \sum_{i=1}^{n} \ln \sin \pi \left(\frac{x_i - a}{d} \right)^s$$

Denote $z_i = \frac{x_i - a}{d}$. Some preliminary calculation:

$$\nabla_s A = \pi k \int_0^1 \sin^{k-1}(\pi z^s) \cos(\pi z^s)(\ln z) z^s \ \mathrm{d}z$$
$$\nabla_k A = \int_0^1 \sin^k(\pi z^s) \ln \sin(\pi z^s) \ \mathrm{d}z$$

Gives us the score equations:

$$\nabla_a \ell = 0 \iff \sum_{i=1}^{n} z_i^{s-1} \cot(\pi z_i^s) = 0; \qquad \nabla_d \ell = 0 \iff \pi s k \sum_{i=1}^{n} z_i^s \cot(\pi z_i^s) = -n$$

$$\nabla_s \ell = 0 \iff \pi k \sum_{i=1}^{n} \frac{z^s \cos \pi z^s \ln z}{\sin \pi z^s} = n \frac{\nabla_s A}{A}; \qquad \nabla_k \ell = 0 \iff \sum_{i=1}^{n} \ln \sin \pi z_i^s = n \frac{\nabla_k A}{A}$$

which may not be solved analytically. It may be wise to abandon this approach in favour of maximizing the joint likelihood through numerical computation means.

6.2 MSM on MSM

Given a set of 4 moment summary measures (MSMs), we focus on the problem of fitting an appropriate *Sinusoidal distribution* to them, so that its MSMs match the specifications as closely as possible. Because $\text{Sinu}(a, d, s, k)$ is a location-scale family in (a, d), implying that its first two MSMs (mean & variance) can be freely adjusted by modifying (a, d) alone, it should be sufficient to investigate the skewness-kurtosis adaptability of the distribution through its parameters s and k. Hence, choosing the Standard $\text{Sinu}(s, k)$ family, we analyze its skewness-kurtosis cover in Figure 5 over a finite grid of parameter choices: $s, k = 0.05\,(0.05)\,6$. The plot is colorized to indicate the choice of parameters as denoted in the legend aside.

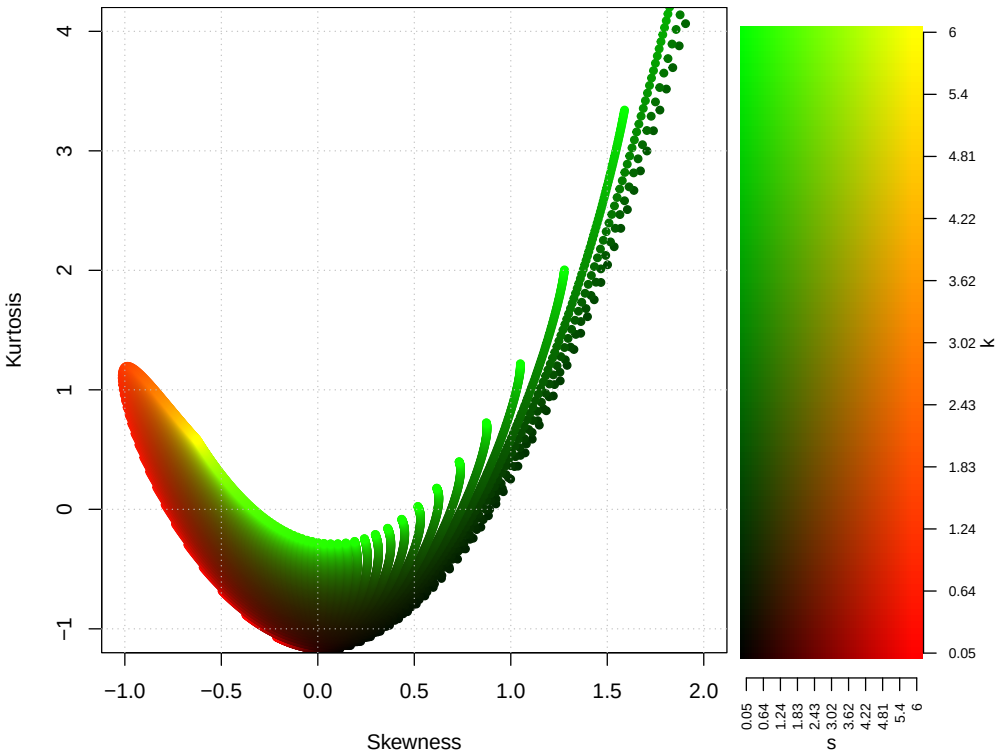


Figure 5: Skewness-Kurtosis cover of the *Sinusoidal distribution*.

At a later stage, we will assess the prospect of optimizing some distance metric over MSMs and

open the path to numerical optimization.

6.3 PDF on PDF

We tried fitting the PDF of $\text{Sinu}(a, d, s, k)$ over different target PDFs (standard location-scale members only) using numerical optimization (`L-BFGS-B` method of the `optim` function found in the `R` programming language) and the loss function as Hellinger distance. For finiteness of computation, we operate under the following parametric constraints whilst choosing the best candidate $\text{Sinu}(a, d, s, k)$: $a \in [-10^3, 10^3]$, $d \in [10^{-2}, 2 \times 10^3]$, $s \in [10^{-2}, 10^2]$ and $k \in [10^{-2}, 10^2]$. The results (minimized Hellinger distance with the choice of parameters) are provided in Table 1.

Target Density	Hellinger distance (Choice of Parameters a, d, s, k)
$\mathcal{N}(0,1)$	0.00004277289 (-20.542932, 34.985498, 1.304535, 100)
$\text{Lognormal}(0,1)$	0.000936801 (0.02827762, 1000, 0.08566970, 35.78972)
$\text{Skew}\mathcal{N}_{\alpha=1}(0,1)$	2.189446e-05 (-7.2216216, 24.8516728, 0.5936823, 100)
$\text{Logistic}(0,1)$	0.003149183 (-80.275639, 102.222513, 2.882431, 100)
$\text{Exp}(1)$	0.008101119 (0.0005708485, 6.5585684475, 0.0621246472, 1.6766166256)
$\text{DE}(0,1)$	0.01599712 (-8.895071, 16.872288, 1.085768, 15.608238)
$\Gamma(0.1,1)$	Poor fit
$\Gamma(100,1)$	Poor fit
$\text{B}(1,1)$	0.0030202, (-0.0007780335, 1.0012066335, 0.1004892336, 0.12190192670)
$\text{Cauchy}(0,1)$	0.06878475 (-126.211809, 149.303341, 4.141315, 100)
$\chi^2(1)$	Poor fit
$\chi^2(10)$	5.791378e-06 (-0.3201950, 97.1337699, 0.2815837, 22.0271292)
$\text{t}_1(0,1)$	0.06878477 (-126.114092, 149.207078, 4.138119, 100)
$\text{t}_{10}(0,1)$	Fit failed
$\text{F}(1,1)$	Poor fit

Table 1: Some common target densities and the best candidate *Sinusoidal distribution*

The technical and theoretical underpinnings behind poor or failed fits will be the matter of future investigation.

6.4 Mode matching

WLoG, assume density is supported on $0, 1$. Steps:

1. Equate $2^{-1/s} =$ given mode. Call the solution s^*.
2. Equate $A(s^*, k) =$ modal density. Call the solution k^*.
3. Optimal density is $\text{Sinu}(s^*, k^*)$. If this optimal density is a poor fit to the target, then either the target is not a Sinusoidal shape, or Sinusoidal distribution approximates is poorly within those bounds.

6.5 CDF on CDF

6.6 CDF on ECDF

6.7 QF on QF

Quantile function on Quantile Function

6.8 QF on EQF

6.9 DQF on DQF

7 On the theory of fitting

7.1 Quantile-Density Based Metrics

Following Staudte 2017, we construct the notion of Divergences based on the density-quantile function $dq(p) = f(F^{-1}(p)) \quad \forall p \in [0, 1]$. The advantages of calculating divergence metrics on this transformed domain is that divergences become location-scale invariant and invariant to "flat" regions in the CDF.

7.2 Divergence Density/Potential

For some target density g, if its divergence $D_g(f)$ from any candidate density f can be represented as the integral

$$\int_{x \in \mathcal{X}} L(f(x), g(x)) \mathrm{d}x$$

then define:

$$V(x, y) = L(y, g(x))$$

as the divergence density/potential associated with g and induced by L. It gives a glimpse at the topography of penalization of $D(\cdot, \cdot)$ with respect to g. This is a useful metric if we want to compare how different tuning parameter α's affect the DPD. The process of fitting a candidate distribution over a given target then becomes merely an act of minimizing the line integral of potential function along the density line whilst ensuring that the area under the density line is 1.

We can conclude that the *Sinusoidal distribution* is indeed more flexible if for different target densities, which may even come from data, it can minimize the divergence better than conventional distributions. At the same time, excess efficiency implies lack of robustness in the presence of contamination. In such cases, we justify the utility of the *Sinusoidal distribution* by saying that it fits to data more flexibly than other distributions, and transparently displays the nature of the underlying divergence metric without interference from the rigidity of its shape.

Almost all divergences are functions of $y - g$, i.e. dependent on the deviation.

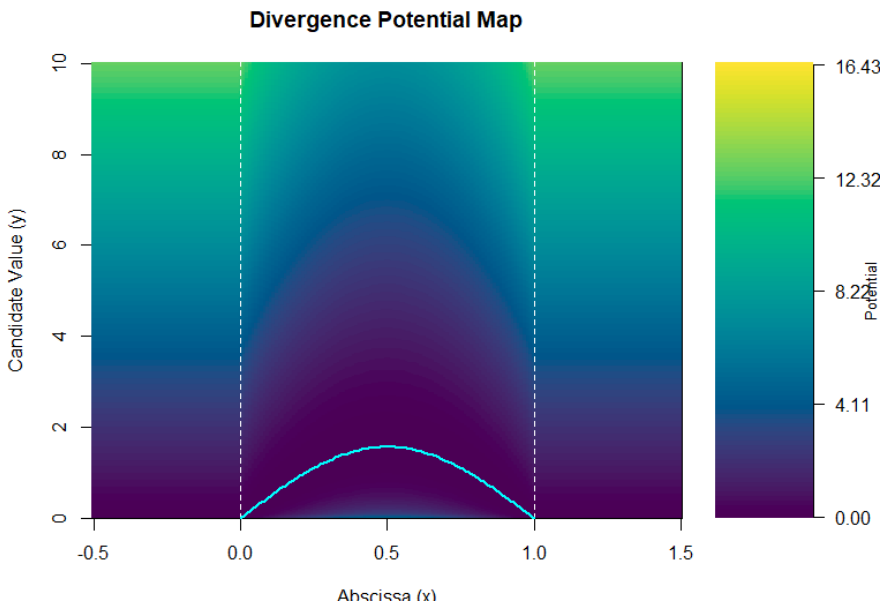


Figure 6: Divergence Potential associated with Sinu$(1, 1)$ and induced by the Hellinger distance

Divergence name	Potential $L_g(y)$	Gradient $\nabla_y L_g(y)$
Hellinger	$\frac{1}{2}(\sqrt{y} - \sqrt{g})^2$	$\frac{1}{2\sqrt{y}}(\sqrt{y} - \sqrt{g})$
DPD	$f^\alpha(f - g) - \frac{1}{\alpha}g(f^\alpha - g^\alpha)$	$(1 + \alpha)y^{\alpha-1}(y - g)$

7.2.1 DPD and contamination

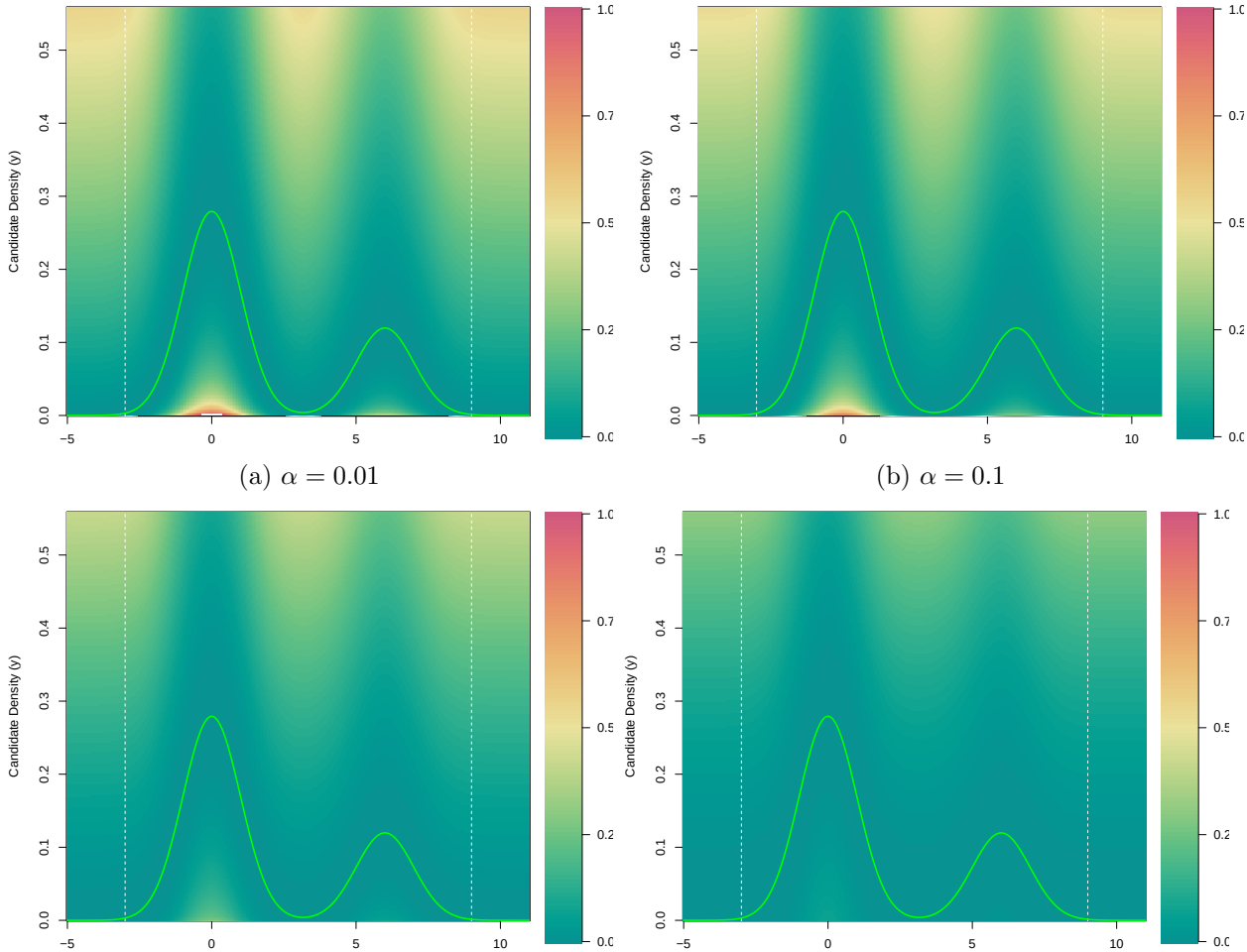
Higher α causes a lower rate of increase of penalization for deviating from true curve, thereby providing more feasibility of a fit that covers both density portions. Whereas a lower α allows rigid maximization over one of the 2 curves only, since passing through the middle high-penalization region in between incurs a lot of cost. See Figure 7.

In the presence of contamination, there is the additional complexity of the principal density being scaled down. So even if the distribution were to fit to the principal density portion, it would it to a dwarved version of it, thereby adjusting a best fit over its "cloud" of minimality. Thereby the candidate distribution is now betrayed to fit over a "non-density" principal component.

Finally observe that for contaminated densities, it is not the "attraction" of the second peak that deviates the candidate to fit contaminatedly, but it is the infeasibility in between that makes unimodal densities suboptimal fits, thereby seeking that the candidate choose either the principal or the contaminant density component.

7.2.2 Calculus of variations

We will work in the DQF domain so as to eliminate the question of location-scale dependence. We are looking at the problem:



(a) $\alpha = 0.01$ (b) $\alpha = 0.1$

(c) $\alpha = 0.5$ (d) $\alpha = 1$

Figure 7: Shape plots of different *Sinusoidal distributions*

$$\inf_{s,k} \int_{y=\psi^*(x;s,k)} L_g(\psi^*(x))\mathrm{d}x$$

$$\inf_{s,k} \int_{\mathcal{C}:y=\psi^*(x;s,k)} \frac{L(y, g(x))}{\sqrt{(1+y')^2}}\, ds$$

Suggestion: A heuristic method to check feasibility of fitting

1. For a contaminated density, choose the local maxima. Call the principal density's maximum x_0, and the subsequent ones $x_1, x_2, \ldots$. We will assume a simple additional unimodal contamination, at x_1.

2. Draw a straight line through the peaks. Find the "deepest gorge" along this line and ascertain the potential at that point. Call it L_{01}. Call the length of the line l_{01}.

3. Maybe put some sort of a threshold on the value of L_{01} and l_{01} based on whether the candidate

density fits principally or contaminatedly.

4. Problems: Not sure if this is a universal approach. Works for unimodal candidate densities, heuristically. Unsure if even then, s, k will spoil the consistency. Or how the separation l_{01} affects the threshold. Ideally, α should not affect the threshold: after all, our task is to ascertain the correct choice of α using the threshold and the graphical details alone of the potential map.

8 Future Plan

In the future, we aim to improve our work in the following departments:

8.1 Analytical replication

We want to investigate parametric limiting cases of the *Sinusoidal distribution* and assess the strength of such convergences. For example, it would be a task to investigate if the *Sinusoidal distribution* converges uniformly to the Normal, Beta, Gamma, etc. distributions in limits of its parameters. Moreover, we would like to investigate what are the families that the *Sinusoidal distribution* cannot replicate – for example, we know through trials that it struggles with leptokurtic shapes. Investigating the theoretical skewness-kurtosis cover of the distribution will also be a priority.

8.2 Analytical expressions

We want to simplify the analytical expressions, such as the CDF, normalizing constant, moments and the characteristic function, of the *Sinusoidal distribution*. For example, we aim to reduce the frequent occurrence of complicated 'double summations' in its analytical expressions by searching for further simplifications.

8.3 Fitting strategies

Firstly, we aim to devise algorithms for fitting the *Sinusoidal distribution* to more statistical objects, viz. sample observations (i.e. formulating a numerical algorithm for maximum likelihood estimation), MSMs, PDFs (formulating and testing algorithms for more divergence metrics), CDFs, and ECDFs. Next, we will compare their efficacy based on a common scheme. Of particular interest in this domain will be location-scale invariant divergences, based on Density-Quantile functions and otherwise.

Following this, we wish to investigate the salient features of different divergences/loss functions by how they allow the *Sinusoidal distribution* to be fitted over different statistical objects. In particular, we would like to understand the influence of the tuning parameter α on the DPD metric, with the ambition of suggesting novel ways of estimating the same, which happens to be a long-standing research question (Basak, Basu, and Jones 2021).

Our argument for the characterization of divergences and their associated (hyper-)parameters by the act of minimizing such divergences during fitting a distribution is that given a flexible distribution (over some variety of shapes), an intuitive way to visualize the impact of different divergence metrics is to see how they influence the fit of the distribution in the minimum divergence estimation approach.

8.4 Test statistic distribution modelling

We want to assess the usage of the distribution to model the distributions of test statistics that do not follow closed form distributions. Because of its highly flexible nature, the *Sinusoidal distribution* is expected to serve as a good proxy distribution for complicated test statistics, against which inference can be done accurately, thereby eliminating the need for elaborate tables in all practical purposes.

8.5 Unconventional datasets

We wish to employ the flexibility of its shape in the analysis of unconventional, skewed or platykurtic, datasets. We hope for applications in the domains of survival analysis and economics, specifically.

8.6 Flexible Bayesian Prior

We want to assess if the additional flexibility of its skewness and kurtosis figures improves the accuracy of Bayesian estimation, by employing it as a highly flexible Bayes prior. We will also assess the computational costs that come with the estimation of 2 additional parameters, especially when they rely on numerical computation, and whether this is an acceptable compromise.

We create an entire linear regression setup based solely on the *Sinusoidal distribution* given its great approximation capabilities, see Mathematical properties as well as empirical results.

$$
\begin{aligned}
\underset{\sim}{Y} &= \mathbf{X}^{\intercal}\underset{\sim}{\beta} + \epsilon_i \\
\epsilon_i &\sim \text{Sinu}(a_Y, d_Y, s_Y, k_Y) \\
\underset{\sim}{\hat{\beta}} &= (\mathbf{X}^{\intercal}\mathbf{X})^{-1}\mathbf{X}^{\intercal}\underset{\sim}{Y} \\
a_Y &\sim \text{Sinu}(a_a, d_a, s_a, k_a) \\
d_Y &\sim \text{Sinu}(a_d, d_d, s_d, k_d) \\
s_Y &\sim \text{Sinu}(a_s, d_s, s_s, k_s) \\
d_Y &\sim \text{Sinu}(a_k, d_k, s_k, k_k)
\end{aligned}
$$

8.7 Flexible Activation Function for NNs

We have already seen in Table 1 that the *Sinusoidal distribution* can take the shape of a Logistic (0,1) distribution with great accuracy. Given the closeness of its CDF to the sigmoid activation function, and the additional ability to take on other shapes as well, we aim to construct a neural network endowed with a flexible skewness-kurtosis activation function. This learnable-parameter NN will require deriving custom back-propagation equations for parameter estimation. We will also assess the additional cost of training such a neural network and the benefits associated with the same.

Input: $a^0_{i_0} \equiv x_i$ for $i \in 1(1)m$
Output: $a^{n+1}_{i_{n+1}} \equiv \hat{y}_i$ for $j \in 1(1)n$
Single-layer network. Middle layer: $z_i = \sum_j w_{ij}x_j + b_i$
Activation: $\sigma(\cdot)$, can be anything.

The neural network is given by:

$$a^{l+1}_{i_{l+1}} = \sigma\left[\sum_{i_l} w_{i_l,i_{l+1}} a^l_{i_l}\right] \quad \begin{cases} \forall i_{l+1} \in 1(1)n_{l+1} \\ \forall l = 0(1)n \end{cases}$$

and $a^l_0, a^{l+1}_0 = 1$ is the bias term at every layer.

Assume Sinunet. Backpropagation equations:

$$f \tag{20}$$

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Appendix

A Code

All code for this project is available at github.com/BoiBhai/Sinusoidal-Distribution.git.

A.1 Choice of Optimizer

Following optimization techniques were tried and their drawbacks realized:

1. `optim`'s `L-BFGS-B` method: sensitivity on initial choices of parameters.
2. `optim`'s `Nelder-Mead` method: unavailability of bounding boxes and the presence of unbounded solutions for certain problems such as $\mathcal{N}(0,1)$ fitting.

B Proofs and Remarks

Theorem B.1. *Mode of* $\mathrm{Sinu}(s,k)$ *is* $2^{-1/s}$ *and the modal density is* $\frac{1}{A(s,k)}$.

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}z}\psi(z;s,k) &= \frac{1}{A(s,k)}\frac{\mathrm{d}}{\mathrm{d}z}\sin^k(\pi z^s) \\
&= \frac{1}{A(s,k)}k\sin^{k-1}(\pi z^s)\cos(\pi z^s)\pi s z^{s-1} \\
&= 0 \iff \begin{cases} \sin(\pi z^s) = 0 \implies z = 0,1 \\ z^{s-1} = 0 \implies z = 0 \\ \cos(\pi z^s) = 0 \implies \pi z^s = \frac{\pi}{2} \implies z = 2^{-1/s} \end{cases}
\end{aligned}$$

And modal density is

$$\psi(2^{-1/s};s,k) = \frac{1}{A(s,k)}\sin^k(\pi z^s)\Big|_{z=2^{-1/s}} = \frac{1}{A(s,k)}$$

□

B.1 Preliminary mathematical properties

Remark B.1 (Physical Interpretation of the *Sinusoidal distribution*). First, note that both the radius of a planet and the angle subtended at its center are actually immaterial with regard to the intensity of crater activity it experiences when we assume that there is no decay in the intensity of crater activity through space, or atleast in the region of space incident normally upon the planet starting from its north-pole to its equator. In fact, what only matters is the tangential inclination of any location on the planet. To summarise, a location $r(z)$ parametrized as above will experience

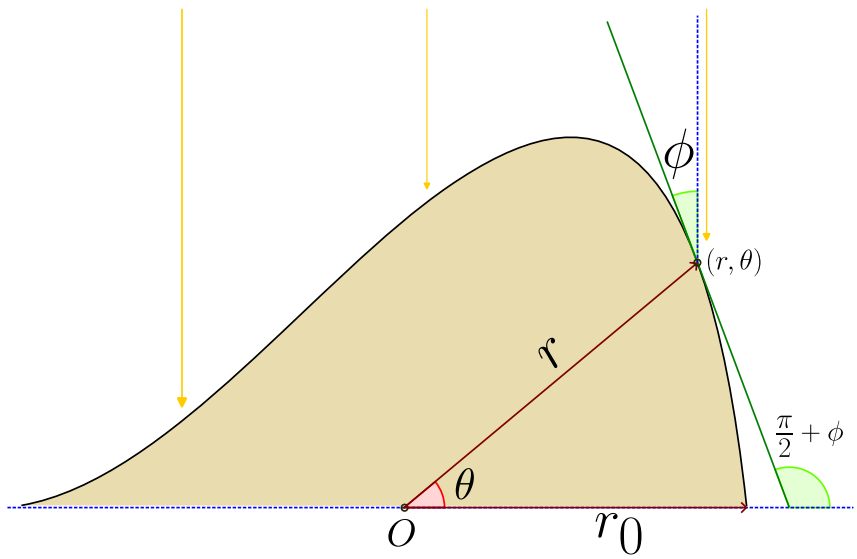

Figure 8: An interpretation of the *Sinusoidal distribution*

activity intensity proportional to $\sin(\phi)$, where ϕ is the angle its tangent subtends against the vertical.

Now, consider a planet whose every location $r(z)$ is inclined at an angle $\phi = \sin^{-1}[\sin^k(\pi z^s)]$ against the vertical. it is immediate that the activity intensity there would be $\sin^k(\pi z^s)$.

The characterization of the shape of such a (smooth) radial curve that defines the planet is a problem in differential geometry. We briefly formulate its solution:

$$x(0) = r_0;\ y(0) = 0$$
$$\phi(z) = \sin^{-1}[\sin^k(\pi z^s)]$$
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\left(\frac{\pi}{2} + \phi\right) \equiv -\cot\phi$$

$$x = r(z)\cos(\pi z^s) \quad \implies \mathrm{d}x = \cos(\pi z^s)\mathrm{d}r - r\sin(\pi z^s)\pi s z^{s-1}\mathrm{d}z$$
$$y = r(z)\sin(\pi z^s) \quad \implies \mathrm{d}y = \sin(\pi z^s)\mathrm{d}r + r\cos(\pi z^s)\pi s z^{s-1}\mathrm{d}z$$

$$
\begin{aligned}
& \frac{\sin(\pi z^s)\mathrm{d}r + r\cos(\pi z^s)\pi s z^{s-1}\mathrm{d}z}{\cos(\pi z^s)\mathrm{d}r - r\sin(\pi z^s)\pi s z^{s-1}\mathrm{d}z} && = -\cot\left[\sin^{-1}\sin^k(\pi z^s)\right] \\
\implies & \frac{\mathrm{d}r}{r} && = \tan\left[\pi z^s - \sin^{-1}\sin^k(\pi z^s)\right]\cdot \pi s z^{s-1}\mathrm{d}z \\
& && = \tan(\theta - \sin^{-1}\sin^k\theta)\mathrm{d}\theta \\
\implies & \ln r\Big|_{r_0}^{r(z)} && = \int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t \\
\implies & r(z) && = r_0 \exp\left(\int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t\right)
\end{aligned}
$$

Proof of Theorem 3.1. Consider the transformation $Y = \sin^2(X)$ for $X \in [0, \pi/2]$. The Jacobian of the transformation is:

$$
\left|\frac{dx}{dy}\right| = \frac{1}{2\sin(x)\cos(x)} = \frac{1}{2y^{1/2}(1-y)^{1/2}} \tag{21}
$$

Substituting the density $f_X(x) \propto \sin^K(x) = y^{K/2}$:

$$
f_Y(y) \propto y^{K/2}\cdot y^{-1/2}(1-y)^{-1/2} = y^{\frac{K+1}{2}-1}(1-y)^{\frac{1}{2}-1} \tag{22}
$$

This identifies Y as Beta $\left(\frac{K+1}{2}, \frac{1}{2}\right)$. The full support $[0, \pi]$ is obtained by reflecting X across $\pi/2$ with probability 0.5 via the Bernoulli variable S.

Let $U = \pi Z^S$. The Jacobian of the inverse transformation $z = (u/\pi)^{1/S}$ is $|dz/du| \propto u^{(1-S)/S}$. The density of U is:

$$
f_U(u) \propto f_Z((u/\pi)^{1/S})\cdot u^{(1-S)/S} \propto (u^{(S-1)/S}\sin^K u)\cdot u^{(1-S)/S} = \sin^K u \tag{23}
$$

on $u \in [0, \pi]$. By the Beta-transformation theorem for $\sin^K u$, the variable U is generated using $Y \sim \mathrm{Beta}(\frac{K+1}{2}, \frac{1}{2})$. Reversing the power transformation $Z = (U/\pi)^{1/S}$ completes the proof. □

Proof of Theorem 3.2.

$$
\begin{aligned}
f(x_1 \ldots x_n) &= \frac{1}{d^n A^n(s,k)}\left[\prod_{j=1}^{n}\sin\pi\left(\frac{x_j - a}{d}\right)^s\right]^k \\
&= \frac{1}{d^n A^n(s,k)}\left[\prod_{j=1}^{n}\sin\theta_j\right]^k
\end{aligned}
$$

Now, the product term can be simplified as:

$$\begin{aligned}
\prod_{j=1}^{n} \sin\theta_j &= \prod_{j=1}^{n} \left(\frac{\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}}{2} \right) \\
&= \frac{1}{2^n} \sum_{r_1=\pm1} \cdots \sum_{r_n=\pm1} \prod_{j=1}^{n} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j} && [\because \text{Claim B.1}] \\
&= \frac{1}{2^n} \sum_{r_1=\pm1} \cdots \sum_{r_n=\pm1} r \exp\left(\mathrm{i} \sum_{j=1}^{n} r_j\theta_j \right) && [\text{let } r := \prod_{j=1}^{n} r_j, \text{ the "product parity"}]
\end{aligned}$$

Plugging the expression back in:

$$f(x_1 \dots x_n) = \frac{1}{A^n(s,k)2^{nk}d^n} \left(\sum_{r_1=\pm1} \cdots \sum_{r_n=\pm1} r \exp\left[\mathrm{i} \sum_{j=1}^{n} r_j\theta_j \right] \right)^k$$

The term $\sum_{r_1=\pm1} \cdots \sum_{r_n=\pm1} r \exp\left[\mathrm{i}\sum_{j=1}^{n} r_j\theta_j\right]$ is of particular interest. Consider n points $z_j = \frac{x_j - a}{d}$ $[j = 1(1)n]$ in $[0,1]$. Construct the angles $\theta_j = \pi z_j^s$ which lie in $[0,\pi]$. Choose a direction vector $\vec{r} = (r_1 \cdots r_n) \in \{\pm1\}^n$, where every direction is $r_i = \pm1$, i.e. either positive or negative. Consider the complex number (corresponding to every choice of a direction vector $\vec{r}$) created by consecutive angular displacements θ_j in the respective directions r_j. Such a complex number is further compensated in sign by the "product parity" r of that direction vector $\vec{r}$. If we sum all such complex numbers created by all possible choices of directions $\vec{r}$, then we get exactly the aforementioned expression. □

Claim B.1.

$$\prod_{j=1}^{n} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) = \sum_{r_1=\pm1} \cdots \sum_{r_n=\pm1} \prod_{j=1}^{n} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}$$

Proof. We prove it by induction. For $n = 2$:

$$\begin{aligned}
\prod_{j=1}^{2} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) &= \mathrm{e}^{\mathrm{i}(\theta_1+\theta_2)} - \mathrm{e}^{\mathrm{i}(\theta_1-\theta_2)} - \mathrm{e}^{\mathrm{i}(\theta_2-\theta_1)} + \mathrm{e}^{-\mathrm{i}(\theta_1+\theta_2)} \\
&= \sum_{r_1\in\{1,-1\}} \sum_{r_2\in\{1,-1\}} r_1 r_2 \cdot \mathrm{e}^{\mathrm{i}r_1\theta_1} e^{\mathrm{i}r_2\theta_2} \\
&\equiv \sum_{r_1,r_2=\pm1} \prod_{j=1}^{2} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}
\end{aligned}$$

Induction hypothesis is $\forall\, n \geq 2$:

$$P(n): \quad \prod_{j=1}^{n} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) = \sum_{r_1=\pm1} \cdots \sum_{r_n=\pm1} \prod_{j=1}^{n} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}$$

We prove it for $n+1$ case:

$$\begin{aligned}
\prod_{j=1}^{n+1}(\mathrm{e}^{\mathrm{i}\theta_j}-\mathrm{e}^{-\mathrm{i}\theta_j}) &= \left(\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1}\prod_{j=1}^{n} r_j\mathrm{e}^{\mathrm{i}r_j\theta_j}\right)\cdot(\mathrm{e}^{\mathrm{i}\theta_{n+1}}-\mathrm{e}^{-\mathrm{i}\theta_{n+1}}) \\
&= \sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1}\prod_{j=1}^{n} r_j\cdot(\mathrm{e}^{\mathrm{i}\theta_{n+1}}-\mathrm{e}^{-\mathrm{i}\theta_{n+1}})\mathrm{e}^{\mathrm{i}r_j\theta_j} \\
&= \sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1}\prod_{j=1}^{n} r_j\cdot\sum_{r_{n+1}=\pm1} r_{n+1}\mathrm{e}^{\mathrm{i}r_{n+1}\theta_{n+1}}\mathrm{e}^{\mathrm{i}r_j\theta_j} \\
&= \sum_{r_1=\pm1}\cdots\sum_{r_{n+1}=\pm1}\prod_{j=1}^{n+1} r_j\mathrm{e}^{\mathrm{i}r_j\theta_j}
\end{aligned}$$

Hence proved. □

Proof of Theorem 3.4.

$$\begin{aligned}
\int_0^1 \sin^k(\pi t)\mathrm{d}t &= \int_0^{\pi}\sin^k\theta\frac{\mathrm{d}\theta}{\pi} \\
&= \frac{2}{\pi}\int_0^{\frac{\pi}{2}}\sin^k\theta\mathrm{d}\theta \\
&= \frac{2}{\pi}\cdot 2^{k-1}\mathrm{B}\left(\frac{k+1}{2},\frac{k+1}{2}\right) && \text{[Zwillinger, Eq. 3.621 (1)]} \\
&\equiv \frac{2^k}{\pi}\frac{\Gamma^2\left(\frac{k+1}{2}\right)}{\Gamma(k+1)}
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\int_0^1 \sin^k(\pi t)\mathrm{d}t &= 2\int_0^{\frac{\pi}{2}}\sin^k\theta\frac{\mathrm{d}\theta}{\pi} \\
&\equiv \frac{2}{\pi}\int_0^{\frac{\pi}{2}}\sin^k\theta\cos^0\theta\mathrm{d}\theta \\
&= \frac{2}{\pi}\cdot\frac{1}{2}\mathrm{B}\left(\frac{k+1}{2},\frac{1}{2}\right) && \text{[Zwillinger, Eq. 3.621 (5)]} \\
&= \frac{1}{\pi}\frac{\Gamma\left(\frac{k+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)} \\
&\equiv \frac{1}{\sqrt{\pi}}\frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)}
\end{aligned}$$

□

Proof of Theorem 3.5.

$$\int_0^z \sin^k(\pi t)\mathrm{d}t = \frac{1}{\pi}\int_0^{\pi z} \sin^k\theta \mathrm{d}\theta \quad \left[\theta := \pi t \implies \mathrm{d}t = \frac{\mathrm{d}\theta}{\pi}\right]$$

$$= \frac{1}{\pi}\cdot \begin{cases} \frac{1}{2^{2n}}\binom{2n}{n}\theta + \frac{(-1)^n}{2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin[2(n-k)\theta]}{2(n-k)} & [k=2n] \\ \frac{1}{2^{2n}}(-1)^{n+1}\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{\cos(2n+1-2k)\theta}{2n+1-2k} & [k=2n+1] \end{cases}\Bigg|_{\theta=0}^{\pi z}$$

[Zwillinger, Eq. 2.513-1, 2]

$$= \frac{1}{\pi}\cdot \begin{cases} \frac{1}{2^{2n}}\binom{2n}{n}\pi z + \frac{(-1)^n}{2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin[2\pi(n-k)z]}{2(n-k)} & [k=2n] \\ \frac{1}{2^{2n}}(-1)^{n}\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{1-\cos(2n+1-2k)\pi z}{2n+1-2k} & [k=2n+1] \end{cases}$$

□

Proof of Theorem 3.6.

$$f(x;d,k) = \frac{1}{dA(1,k)}\sin^k\left[\pi\frac{x+\frac{d}{2}}{d}\right]$$

$$= \frac{\sqrt{\pi}\cdot\Gamma(k+1)}{d\cdot\Gamma\left(\frac{k+1}{2}\right)}\cos^k\left(\frac{\pi x}{d}\right) \qquad \text{[Equation 8]}$$

$$\sim \frac{\sqrt{\pi}\cdot k^{k+\frac{1}{2}}\mathrm{e}^{-k}}{d\cdot\left(\frac{k-1}{2}\right)^{k/2}\mathrm{e}^{-\frac{k-1}{2}}}\exp\left[k\ln\cos\left(\frac{\pi x}{d}\right)\right] \qquad \text{[Stirling approximation for large } k\text{]}$$

$$\sim \frac{\sqrt{\pi}\cdot k^{k+\frac{1}{2}}\left(\frac{\mathrm{e}}{2}\right)^{-\frac{k}{2}}}{d\cdot k^{k/2}}\exp\left[k\ln\left(1-\frac{\pi^2x^2}{2d^2}+\mathcal{O}(d^{-4})\right)\right] \qquad \text{[Maclaurin series in } k \text{ for cos]}$$

$$\sim \frac{\sqrt{\pi}}{d}k^{\frac{k+1}{2}}\left(\frac{\mathrm{e}}{2}\right)^{-\frac{k}{2}}\exp\left[-\frac{\pi^2kx^2}{2d^2}+k\mathcal{O}(d^{-4})\right] \qquad \text{[Mercator series for ln]}$$

Let us set $d = \pi\sqrt{k}$, so that:

$$f(x;d,k) \sim \frac{1}{\sqrt{\pi}}k^k\left(\frac{\mathrm{e}}{2}\right)^{-\frac{k}{2}}\exp\left[-\frac{x^2}{2}+k\mathcal{O}(k^{-2})\right]$$

$$\propto \exp\left[-\frac{x^2}{2}+\mathcal{O}(k^{-1})\right]$$

$$\to \mathcal{N}(0,1) \qquad \text{as } k\to\infty$$

□

B.2 Properties involving the GHGF

Remark B.2 (Additional notes on GHGFs). For simplicity, all HGFs (e.g. Confluent HGF) expressed in this paper are represented in the Generalized form only.

Note than if $\mathrm{F}_j := \frac{(a_1)^{\overline{j}}\cdots(a_p)^{\overline{j}}}{(b_1)^{\overline{j}}\cdots(b_q)^{\overline{j}}}\frac{x^j}{j!}$ denotes the jth term ($j = 0(1)\infty$) of the summation, so that

$\mathrm{F}_q^p \equiv \sum_{j=0}^{\infty} \mathrm{F}_j$, then the ratio of consecutive terms of a GHgF defines its parameters and arguments:

$$\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n}\left[\begin{matrix} a_1 \cdots a_p \\ b_1 \cdots b_1 \end{matrix}; z\right] = \frac{(a_1+n)\cdots(a_p+n)}{(b_1+n)\cdots(b_q+n)}\frac{z}{n+1}$$

In our paper, we will also need the Generalized Binomial expansion formula because k may not be integer. Define from hereon $\binom{r}{k} = \frac{(r)^{\overline{k}}}{k!}$ as the generalized binomial coefficient. Then $(x+y)^r = \sum_{k=0}^{\infty}\binom{r}{k}x^{r-k}y^k$, which is the same series as the finite expansion obtained for $r \in \mathbb{Z}^+$, and otherwise an infinite series.

Proof of Theorem 4.1.

$$\begin{aligned}
\sin^k(\pi z^s) &= \sin^k\theta && [\text{let } \theta := \pi z^s] \\
&= \left(\frac{\mathrm{e}^{i\theta} - \mathrm{e}^{-\mathrm{i}\theta}}{2i}\right)^k && \because \mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta \\
&= \frac{1}{(2\mathrm{i})^k}\sum_{n=0}^{\infty}\binom{k}{n}\mathrm{e}^{\mathrm{i}\theta(k-n)}(-\mathrm{e})^{-\mathrm{i}\theta n} \\
&= \frac{1}{(2\mathrm{i})^k}\sum_{n=0}^{\infty}(-1)^n\binom{k}{n}\mathrm{e}^{\mathrm{i}\theta(k-2n)} \\
&\equiv \frac{1}{(2\mathrm{i})^k}\sum_{n=0}^{\infty}(-1)^n\binom{k}{n}\mathrm{e}^{\mathrm{i}(k-2n)\pi z^s}
\end{aligned}$$

□

Proof of Theorem 4.2. For the first expression, we work with the Maclaurin series of $\sin\theta$ in θ:

$$\begin{aligned}
\sin^k(\pi u) &= (\pi u)^k\left[\sum_{j=0}^{\infty}\frac{(-1)^j(\pi u)^{2j}}{(2j+1)!}\right]^k \\
&= (\pi u)^k\left[\sum_{j=0}^{\infty}\frac{\left(-\frac{\pi^2u^2}{4}\right)^j}{\left(\frac{3}{2}\right)^{\overline{j}}j!}\right]^k && \because \left(\frac{3}{2}\right)^{\overline{j}} = \frac{(2j+1)!}{2^{2j}j!} \\
&= (\pi u)^k\left(\mathrm{F}_1^0\left[\begin{matrix}\cdot \\ \frac{3}{2}\end{matrix}; -\frac{\pi^2u^2}{4}\right]\right)^k
\end{aligned}$$

Alternatively, we construct the required HGF from the ratio of its terms, taking inspiration from Remark B.2. Note that if we were to express the infinite summation in Equation 11 as a HGF, then the ratio of its terms would give:

$$\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n} = (k+n)\frac{-\exp(-2\mathrm{i}\pi z^s)}{n+1}$$

This should be equivalent to the following HGF:

$$\mathrm{F}_0^1\left[\begin{matrix}k\\ \cdot\end{matrix}; -\exp(-2\mathrm{i}\pi z^s)\right]$$

We find that a dual representation is possible, but not much reconciliation or interpretation to this duality is found as of yet. □

Proof of Theorem 4.3.

$$\begin{aligned}\varphi_Z(t) &= \int_0^\infty \mathrm{e}^{\mathrm{i}tz}\psi(z;s,k)\mathrm{d}z = \frac{1}{A(s,k)}\int_0^\infty \mathrm{e}^{\mathrm{i}tz}\varsigma(z;s,k)\mathrm{d}z \\ &= \frac{1}{A(s,k)}\int_0^1 \mathrm{e}^{\mathrm{i}tz}\frac{1}{(2i)^k}\sum_{j=0}^\infty \binom{k}{j}(-1)^j\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{[Equation 11]} \\ &= \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^\infty(-1)^j\binom{k}{j}\int_0^1 \mathrm{e}^{\mathrm{i}tz}\mathrm{e}^{\pi(k-2j)z^s}\mathrm{d}z \end{aligned} \tag{24}$$

We evaluate the inner integral $\mathcal{I}_1 := \int_0^1 \mathrm{e}^{\mathrm{i}tz}\mathrm{e}^{\pi(k-2j)z^s}\mathrm{d}z$ as follows:

$$\begin{aligned}\mathcal{I}_1 &= \int_0^1\left(\sum_{m=0}^\infty \frac{(\mathrm{i}t)^m z^m}{m!}\right)\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{[Maclaurin series for } \mathrm{e}^{\mathrm{i}tz} \text{ in } z\text{]} \\ &= \sum_{m=0}^\infty \frac{(\mathrm{i}t)^m}{m!}\int_0^1 z^m\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \left[\text{Exchanging } \sum \text{ and } \int\right] \end{aligned} \tag{25}$$

We again evaluate the inner integral $\mathcal{I}_2 := \int_0^1 z^m\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z$, by writing $\omega := \pi(k-2j)$ for convenience and substituting $u = z^s$, thereby $\implies z = u^{1/s}$ and $\mathrm{d}z = \frac{1}{s}u^{1/s-1}\mathrm{d}u$:

$$\begin{aligned}\mathcal{I}_2 &= \int_0^1 (u^{1/s})^m\mathrm{e}^{\mathrm{i}\omega u}\frac{1}{s}u^{1/s-1}\mathrm{d}u \\ &= \frac{1}{s}\int_0^1 u^{\frac{m+1}{s}-1}\mathrm{e}^{i\omega u}\mathrm{d}u \\ &= \frac{1}{s}\frac{1}{\frac{m+1}{s}}\mathrm{F}_1^1\left[\begin{matrix}\frac{m+1}{s}\\ \frac{m+1}{s}+1\end{matrix}; \mathrm{i}\omega\right] && \text{NIST DLMF, 13.2.E4, 13.4.E1} \end{aligned} \tag{26}$$

Hence, substituting the formulations Equation 25, Equation 26 back into the parent equation Equation 24, we have –

$$\varphi_Z(t) = \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^\infty(-1)^j\binom{k}{j}\sum_{m=0}^\infty\frac{(\mathrm{i}t)^m}{m!(m+1)}\mathrm{F}_1^1\left[\begin{matrix}\frac{m+1}{s}\\ \frac{m+1}{s}+1\end{matrix}; \mathrm{i}\pi(k-2j)\right]$$

which completes the proof. □

Proof of Theorem 4.4. Since it is known that $E[Z^r] = \frac{1}{\mathrm{i}^r}\left[\frac{\mathrm{d}^r}{\mathrm{d}t^r}\varphi_Z(t)\right]_{t=0}$, by r^{th} order differen-

tiation of the Characteristic Function in Equation 14, we obtain:

$$\mu_r(s,k) = \frac{1}{\mathrm{i}^r} \cdot \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} \frac{(-1)^j \binom{k}{j} (\mathrm{i}^r r!)}{(r+1)!} \, \mathrm{F}_1^1 \left[\begin{matrix} \frac{r+1}{s} \\ \frac{r+1}{s} + 1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right]$$

$$\left[\because \left. \frac{\mathrm{d}^r}{\mathrm{d}t^r} (\mathrm{i}t)^m \right|_{t=0} = \begin{cases} \mathrm{i}^r r! & \text{if } m = r \\ 0 & \text{o.w.} \end{cases} \right]$$

This proves the first expression. Alternatively, by direct integration of the PDF:

$$\begin{aligned} \int z^r \sin^k(\pi z^s) \mathrm{d}z &= \int z^r \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s} \mathrm{d}z && [\because \text{Equation 11}] \\ &= \frac{1}{(2\mathrm{i})^k} \int \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi z^s]^n}{n!} z^r \mathrm{d}z && [\text{Maclaurin series for } \mathrm{e}^\theta \text{ in } \theta] \\ &= \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!} \int z^{ns+r} \mathrm{d}z && \left[\text{Substitute } \int \text{ and } \sum\right] \end{aligned} \tag{27}$$

$$\therefore \mu_r(s,k) = \frac{1}{A(s,k)} \int_0^1 z^r \sin^k(\pi z^s) \mathrm{d}z = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+r+1)}$$

Note that the second summation is precisely the term F_1^1. □

Proof of Theorem 4.5. This follows from the calculation of moments in Theorem 4.4:

$$\begin{aligned} \Psi(z;s,k) &= \frac{1}{A(s,k)} \int_0^z \sin^k(\pi t^s) \mathrm{d}t \\ &= \frac{1}{A(s,k)} \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} \binom{k}{j} (-1)^j \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!} \int_0^z t^{ns} \mathrm{d}t && [\because \text{Equation 27, putting } r=0] \\ &= \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)} z^{ns+1} \end{aligned}$$

□